Mock Modularity In CHL Models

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Outline

Black Hole Microstate Counting

Background

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- One of the great successes of string theory: microscopic description of black holes. [Strominger-Vafa (1996), Dijkgraaf-Verlinde-Verlinde (1996), Shih-Strominger-Yin (2005), Gaiotto (2005), Shih-Yin (2005), Jatkar-Sen (2005), David-Jatkar-Sen (2006)].

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Find the Bekenstein-Hawking entropy of $\frac{1}{4}$-BPS extremal black holes carrying electric and magnetic charges $(\vec{Q}, \vec{P})$ in Type IIB supergravity compactified on $\mathcal{M} \times S^1 \times \widetilde{S}^1$ with $\mathcal{M} = K3, T^4$.

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- Perfect match at leading order! [Reviews: Sen (2007), Mandal-Sen (2010)]

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- Goal: generalize the story of DMZ to more general string compactifications.

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- $N = 1$ with $\mathcal{M} = K3$ is the DMZ case.

Degeneracy of $\frac{1}{4}$-BPS Dyons

Degeneracy formula [Dijkgraaf-Verlinde-Verlinde (1994), David-Jatkar-Sen (2006)]

$$d(\vec{Q},\vec{P}) = \frac{(-1)^{\ell+1}}{N}\int_{\mathcal{C}} d\tau d\sigma dz\, \frac{e^{-2\pi i(m\tau+n\sigma+\ell z)}}{\widetilde{\Phi}_k(\tau,z,\sigma)}$$

where $\widetilde{\Phi}_k$ is a Siegel modular form of weight k for some subgroup of $\mathrm{Sp}(2,\mathbb{Z})$, the contour $\mathcal{C}$ is given by

$$\mathcal{C}: \mathrm{Im}(\tau) = \frac{P^2}{N\varepsilon} \equiv \frac{2n}{N\varepsilon}, \quad \mathrm{Im}(\sigma) = \frac{NQ^2}{\varepsilon} \equiv \frac{2m}{\varepsilon},$$
$$\mathrm{Im}(z) = -\frac{P\cdot Q}{\varepsilon} \equiv -\frac{\ell}{\varepsilon},$$
$$0 \le \mathrm{Re}(\tau), \mathrm{Re}(z) < 1,\ 0 \le \mathrm{Re}(\sigma) < N\ ,$$

and $\varepsilon > 0$ is a small real number.

Some Number Theory Background

Modular Forms Of Integral Weight

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A holomorphic function $f : \mathbb{H} \to \mathbb{C}$ is called a modular form of weight k for $\Gamma \subset \mathrm{SL}(2, \mathbb{Z})$ if

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k f(\tau), \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma \ .$$

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Fourier expansion at $i\infty$

$$f(\tau) = \sum_{n \in \mathbb{Z}} a_f(n) q^{n/h} \ ,$$

where $h > 0$ is the smallest integer such that $\left(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix}\right) \in \Gamma$.

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where $\left(\frac{c}{d}\right)$ is the Jacobi symbol, for odd d

$$\varepsilon_d = \begin{cases} 1 & \text{if } d \equiv 1 \bmod 4; \\ i & \text{if } d \equiv -1 \bmod 4 \ , \end{cases}$$

and

$$\Gamma_0(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}) : c \ \equiv \ 0 \ (\mathrm{mod}\ N) \right\} \ .$$

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The vector space of modular forms of weight $\kappa \in \frac{1}{2}\mathbb{Z}$ is denoted by $M_\kappa(\Gamma)$.

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$$\varphi\left(\frac{a\tau+b}{c\tau+d}, \frac{z+\lambda\tau+\mu}{c\tau+d}\right)$$
$$= (c\tau+d)^k \mathbf{e}\left(m\left(\frac{c(z+\lambda\tau+\mu)^2}{c\tau+d} - \lambda^2\tau - 2\lambda z - \lambda\mu\right)\right)\varphi(\tau,z),$$
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The space of such forms is denoted by $J^{\mathrm{mer}}_{k,m}(\Gamma \ltimes E)$.

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Symplectic group

$$\mathrm{Sp}(2, \mathbb{R}) = \{ M \in \mathrm{GL}(4, \mathbb{R}) : M^t J M = J, \ J := \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix} \},$$

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A Siegel modular form of weight $k \in \mathbb{Z}$ for $G \subset \mathrm{Sp}(2,\mathbb{Z})$ is a meromorphic function $\Phi : \mathbb{H}_2 \to \mathbb{C}$ such that for all $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \in G$ we have

$$\Phi((A\Omega + B)(C\Omega + D)^{-1}) = \mathrm{Det}(C\Omega + D)^k \Phi(\Omega).$$

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$$\Phi(\tau, z, \sigma) = \sum_{n,m,r \in \mathbb{Z}} a(n, m, r) q^{n/N_1} \zeta^{r/N_2} p^{m/N_3} \ ,$$

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where $q = \mathbf{e}(\tau), \zeta = \mathbf{e}(z), p = \mathbf{e}(\sigma)$ and $\mathbf{e}(x) = e^{2\pi i x}$. The sum over indices n, m, r may be restricted depending on the analytic properties of Φ.

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This is called the Fourier-Jacobi expansion for the following reason: using the embedding of $\mathrm{SL}(2,\mathbb{Z})$ and $\mathbb{Z}^2$ into $\mathrm{Sp}(2,\mathbb{Z})$ given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a & 0 & b & 0 \\ 0 & 1 & 0 & 0 \\ c & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (\lambda, \mu) \mapsto \begin{pmatrix} 1 & 0 & 0 & \lambda \\ -\mu & 1 & \lambda & 0 \\ 0 & 0 & 1 & \mu \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

one can show that each φ_m transforms like a Jacobi form of weight k and index $\frac{m}{N_3}$ with respect to some subgroup of $\mathrm{SL}(2,\mathbb{Z}) \ltimes \mathbb{Z}^2$.

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$$\vartheta_{m,\ell}(\tau, z) := \sum_{\substack{r \in \mathbb{Z} \\ r \equiv \ell (\mathrm{mod}\ 2m)}} q^{r^2/4m} \zeta^r = \sum_{n \in \mathbb{Z}} q^{(\ell+2mn)^2/4m} \zeta^{\ell+2mn}.$$

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Then each $h_\ell \in M_{k-\frac{1}{2}}(\Gamma(4mN))$ where

$$\Gamma(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}) : \begin{array}{c} a \equiv d \equiv 1 (\mathrm{mod}\ N) \\ b \equiv c \equiv 0 (\mathrm{mod}\ N) \end{array} \right\} .$$

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A weight $\kappa \in \frac{1}{2}\mathbb{Z}$ mock modular form is a holomorphic function $f : \mathbb{H} \to \mathbb{C}$ such that there exists $g \in M_{2-\kappa}(\Gamma)$ such that $f + g^*$ transforms like a weight-κ modular form with respect to Γ, where

$$g^*(\tau) = 2^{\kappa-1} i \int_{-\bar{\tau}}^{i\infty} d\omega \frac{\overline{g(-\bar{\omega})}}{(-i(\omega+\tau))^\kappa} \ ,$$

is the *Eichler integral* of g.

Mock Modular Forms

Idea:

- Holomorphic but not modular.
- Can be completed to a modular form by adding a nonholomorphic function.
- The completion is necessarily nonholomorphic.

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is the *Eichler integral* of g. The modular form g is called the *shadow* of f and $f + g^*$ is called the *completion* of f.

Mock Jacobi Forms

Mock Jacobi Forms

Mock Jacobi Form

A holomorphic function

$$\varphi(\tau, z) = \sum_{\ell \bmod 2m} h_\ell(\tau)\vartheta_{m,\ell}(N\tau, z) \ ,$$

is called a mock Jacobi form of weight k and index $\frac{m}{N} \in \mathbb{Z}/N$ if each h_ℓ is a weight-$(k-1/2)$ mock modular form for some subgroup of $\mathrm{SL}(2,\mathbb{Z})$ and the completion

$$\widehat{\varphi}(\tau, z) = \sum_{\ell \bmod 2m} (h_\ell(\tau) + g_\ell^*(\tau))\vartheta_{m,\ell}(N\tau, z)$$

transforms like a Jacobi form of weight k and index $\frac{m}{N} \in \mathbb{Z}/N$.

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and B_k are Bernoulli numbers defined by $\frac{x}{e^x - 1} = \sum_{k=0}^{\infty} B_k \frac{x^k}{k!}$.

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$$M_k(\mathrm{SL}(2,\mathbb{Z})) = \bigoplus_{\substack{4\ell_1 + 6\ell_2 = k \\ \ell_1, \ell_2 \geq 0}} \mathbb{C} \cdot E_4^{\ell_1} E_6^{\ell_2} \ .$$

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Another example: Ramanujan discriminant function (weight 12)

$$\Delta(\tau) = \frac{E_4(\tau)^3 - E_6(\tau)^2}{1728} = q \prod_{n=1}^{\infty} (1 - q^n)^{24} \ .$$

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We also have $\eta^{24} = \Delta$.

Examples: Jacobi forms

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Jacobi theta functions

$$\vartheta_1(\tau,z)=\sum_{n\in\mathbb{Z}}(-1)^n q^{\frac{1}{2}(n-\frac{1}{2})^2}\zeta^{n-\frac{1}{2}},\quad \vartheta_2(\tau,z)=\sum_{n\in\mathbb{Z}} q^{\frac{1}{2}(n-\frac{1}{2})^2}\zeta^{n-\frac{1}{2}},$$

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Define $\varphi_{k,1}$ by

$$\varphi_{-2,1}(\tau,z):=\frac{\vartheta_1^2(\tau,z)}{\eta^6(\tau)},\quad \varphi_{10,1(\tau,z)}:=\eta^{24}(\tau)\varphi_{-2,1}(\tau,z)\ ,$$

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where $\vartheta_i(\tau)=\vartheta_i(\tau,0)$.

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where $\vartheta_i(\tau)=\vartheta_i(\tau,0)$. Then $\varphi_{k,1}\in J_{k,1}(\mathrm{SL}(2,\mathbb{Z})\ltimes\mathbb{Z}^2)$. In fact

$$J_{k,m}(\mathrm{SL}(2,\mathbb{Z})\ltimes\mathbb{Z}^2)=\bigoplus_{j=0}^{m} M_{k+2j}(\mathrm{SL}(2,\mathbb{Z}))\cdot A^j B^{m-j}\ ,$$

where $A=\varphi_{-2,1},\ B=\varphi_{0,1}$.

Mock Modular Forms And Mock Jacobi Forms

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Generating function of Hurwitz class numbers is a mock modular form of weight $3/2$ for $\Gamma_0(4)$ with shadow $-\frac{1}{16}\Theta$ [Zagier (1975)]:

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where $H(0) = -1/12$. Then

$$\mathbf{H}(\tau, z) = \mathbf{H}_0(\tau)\vartheta_{1,0}(\tau, z) + \mathbf{H}_1(\tau)\vartheta_{1,1}(\tau, z) = \sum_{\substack{n,r\in\mathbb{Z} \\ 4n-r^2\geq 0}} H\left(4n-r^2\right) q^n \zeta^r$$

is a mock Jacobi form of weight 2 and index 1 [DMZ (2012)].

Meromorphicity and Mock Modularity

Dabholkar-Murthy-Zagier Result

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DMZ Result

Let $\varphi(\tau, z) \in J^{\mathrm{mer}}_{k,m}(\mathrm{SL}(2,\mathbb{Z}) \ltimes \mathbb{Z}^2)$ with single or double poles at $z_s = \alpha_s \tau + \beta_s \in \mathbb{Q}\tau + \mathbb{Q}$. Then there is a decomposition

$$\varphi = \varphi^F + \varphi^P \ ,$$

where φ^F (called the finite part) is a holomorphic (mixed) mock Jacobi form and φ^P (called polar part) has the same pole structure as φ but is not modular.

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- Easily generalizes to $\varphi(\tau, z) \in J^{\text{mer}}_{k,m}(\Gamma \ltimes \mathbb{Z}^2)$ for subgroups $\Gamma \subset \mathrm{SL}(2,\mathbb{Z})$.

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- In CHL models, we encounter meromorphic Jacobi forms in $J^{\text{mer}}_{k,\frac{m}{N}}(\Gamma \ltimes (N\mathbb{Z} \times \mathbb{Z}))$ with $\Gamma = \Gamma_0(N),\ \Gamma_1(N)$ with a double pole at $z = 0 \in \mathbb{Q}\tau + \mathbb{Q}$.

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Let $\varphi(\tau, z) \in J^{\text{mer}}_{k,m}(\mathrm{SL}(2,\mathbb{Z}) \ltimes \mathbb{Z}^2)$ with single or double poles at $z_s = \alpha_s \tau + \beta_s \in \mathbb{Q}\tau + \mathbb{Q}$. Then there is a decomposition

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- The goal is to generalize DMZ result to such Jacobi forms.
- Taking $N = 1$ reduces to DMZ case.

Generalizing DMZ

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- **Step 1:** Start with $\psi(\tau, z) \in J^{\text{mer}}_{k,\frac{m}{N}}(\Gamma \ltimes \mathbb{Z}^2)$ with $\Gamma = \Gamma_0(N), \Gamma_1(N)$ and construct ψ^P which is not modular and $\psi^F := \psi - \psi^P$ is holomorphic.

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$$\psi(\tau, z+\lambda\tau+\mu) = \mathbf{e}\left(-m\left(\lambda^2\tau + 2\lambda z + \lambda\mu\right)\right)\psi(\tau, z), \quad (\lambda, \mu) \in N\mathbb{Z}\times\mathbb{Z},$$

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Strategy: Find functions $\tilde{\mathcal{A}}^s_{\frac{m}{N}}, \tilde{\mathcal{B}}^s_{\frac{m}{N}}$ which have simple and double poles with residue 1 at $z = z_{s+(\lambda,\mu)}$ respectively. Multiply these functions with the residue of ψ to get ψ^P.

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Define the functions at $z = z_s$ by

$$\mathbf{e}\left(\frac{m}{N}\alpha(z_s + 2\varepsilon)\right)\psi\left(\tau, z_s + \varepsilon\right) = \frac{E_s(\tau)}{(2\pi i\varepsilon)^2} + \frac{D_s(\tau)}{2\pi i\varepsilon} + O(1), \quad \varepsilon \to 0\ .$$

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Start with [DMZ (2012)]

$$\mathcal{R}_c^{(1)}(\mathbf{e}(z)) = \frac{1}{2\pi i}\sum_{n\in\mathbb{Z}} \frac{\mathbf{e}(c(z-n))}{z-n}, \quad \mathcal{R}_c^{(2)}(\mathbf{e}(z)) = \frac{1}{(2\pi i)^2}\sum_{n\in\mathbb{Z}} \frac{\mathbf{e}(c(z-n))}{(z-n)^2},$$

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Generalizing DMZ: Step 1 Continued...

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Apply the averaging operator and define *universal Appell-Lerche sums*

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$$\tilde{\mathcal{A}}^s_{\frac{m}{N}}(\tau,z) := \mathbf{e}\left(-\frac{m}{N}\alpha z_s\right)\sum_{r\in\mathbb{Z}} q^{mNr^2}\zeta^{2mr}\mathcal{R}^{(1)}_{-2m\alpha/N}\left(\frac{q^{Nr}\zeta}{\zeta_s}\right)$$

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Define the polar part by

$$\psi^P(\tau,z) = \sum_{s\in S(\psi)/(N\mathbb{Z}\times\mathbb{Z})}\left(D_s(\tau)\tilde{\mathcal{A}}^s_m(\tau,z) + E_s(\tau)\tilde{\mathcal{B}}^s_m(\tau,z)\right)\ ,$$

where $S(\psi)\subset\mathbb{Q}^2$ is the set of poles of ψ.

Generalizing DMZ: Step 3

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where

$$h_\ell(\tau) = q^{-\ell^2 N/4m} \int_{-\ell N\tau/2m}^{-(\ell N\tau/2m)+1} \psi(\tau, z')\mathbf{e}(-\ell z')dz' \ .$$

ψ^F is a (mixed) mock Jacobi form of weight k, index m/N for Γ. The completion of $h_\ell(\tau)$ is given by

$$\widehat{h}_\ell(\tau) = h_\ell(\tau) - \sum_{s\in S(\psi)/(N\mathbb{Z}\times\mathbb{Z})} \left(D_s(\tau)\widetilde{\Theta}^{s*}_{\frac{m}{N},\ell}(\tau) + E_s(\tau)\widetilde{\Xi}^{s*}_{\frac{m}{N},\ell}(\tau)\right) \ ,$$

Generalizing DMZ: Step 3 Continued...

Mock Modularity of ψ^F [Bhand-Sen-RKS (2023)]

where $\widetilde{\Theta}^{s*}_{\frac{m}{N},\ell}, \widetilde{\Xi}^{s*}_{\frac{m}{N},\ell}(\tau)$ are Eichler integrals of

$$\widetilde{\Theta}^{s}_{\frac{m}{N},\ell}(\tau) = \mathbf{e}\left(-\frac{m}{N}\alpha\beta\right) \sum_{r\in\mathbb{Z}+\frac{\alpha}{N}+\frac{\ell}{2m}} r\ \mathbf{e}(2m\beta r) q^{Nmr^2}\ ,$$

$$\widetilde{\Xi}^{s}_{\frac{m}{N},\ell}(\tau) = \mathbf{e}\left(-\frac{m}{N}\alpha\beta\right) \sum_{r\in\mathbb{Z}+\frac{\alpha}{N}+\frac{\ell}{2m}} \mathbf{e}(2m\beta r) q^{Nmr^2}\ ,$$

respectively.

Mock Modularity In CHL Models

Dyon Partition Function In CHL Models

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$$k = \begin{cases} \frac{24}{N+1} - 2, & N = 1,2,3,5,7; \\ 3, & N = 4; \\ 2, & N = 6; \\ 1, & N = 8. \end{cases}$$

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For $\mathcal{M} = K3,\ N = 1$ (DMZ case), the partition function $\widetilde{\Phi}_k^{-1} = \Phi_{10}^{-1}$ is the inverse of the well-known Igusa cusp form.

Poles Of $\widetilde{\Phi}_k^{-1}$

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Zeros of $\widetilde{\Phi}_k$ [David-Jatkar-Sen (2006)]

The partition function $\widetilde{\Phi}_k$ has double zeros on the hypersurface

$$n_2\left(\sigma\tau - z^2\right) + jz + n_1\sigma - m_1\tau + m_2 = 0$$

where

$$m_1 \in N\mathbb{Z}, \quad n_1 \in \mathbb{Z}, \quad j \in 2\mathbb{Z}+1, \quad m_2, n_2 \in \mathbb{Z}, \quad m_1n_1+m_2n_2+\frac{j^2}{4} = \frac{1}{4}.$$

The behaviour near $z = 0$ is

$$\widetilde{\Phi}_k(\tau, z, \sigma) = -4\pi^2 z^2 f^{(k)}(N\tau) g^{(k)}(\sigma/N) + O(z^4),$$

for some modular forms $f^{(k)}, g^{(k)}$.

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Fourier-Jacobi expansion of $\widetilde{\Phi}_k^{-1}$ [Bhand-Sen-RKS (2023)]

$$\frac{1}{\widetilde{\Phi}_k(\tau,z,\sigma)} = \sum_{m=-\widetilde{\alpha}}^{\infty} \phi_m(\sigma,z) q^m, \quad q=\mathbf{e}(\tau), \quad \mathrm{Im}(\tau)\to\infty$$

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- The finite parts ϕ_m^F, ψ_n^F are mock Jacobi forms of appropriate type.

Degeneracy Of Single-Centered Black Holes

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For charges $Q^2 = \frac{2m}{N}, P^2 = 2n, Q \cdot P = \ell$ with $(n, \ell, m) \in \mathbb{Z}^3$,

$$d(\vec{Q}, \vec{P}) = \frac{(-1)^{\ell+1}}{N} \int_{\mathcal{C}} d\tau d\sigma dz \, \frac{e^{-2\pi i(m\tau + n\sigma + \ell z)}}{\widetilde{\Phi}_k(\tau, z, \sigma)}$$

with $\mathcal{C} : \mathrm{Im}(\tau) = \frac{2n}{N\varepsilon}, \quad \mathrm{Im}(\sigma) = \frac{2m}{\varepsilon}, \quad \mathrm{Im}(z) = -\frac{\ell}{\varepsilon}$ and

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Contour Deformation [Bhand-Sen-RKS (2023)]

(1) Deformation of $\mathcal{C}$ from $\mathrm{Im}(\tau) = \frac{2n}{N\varepsilon}$ to $\mathrm{Im}(\tau) \to \infty$ does not get contribution from poles of $\widetilde{\Phi}_k^{-1}$ if and only if $(n, \ell, m) \in A$ where $A \subset \mathbb{Z}^3$ is a set of triples (n, ℓ, m) with $n, m > 0, 4mn - N\ell^2 > 0$ and

$$\left[\left(\frac{m_1\ell + 2ms + m}{m}\right)^2 + \frac{m_1^2}{m^2}\left(4\frac{n}{N}m - \ell^2\right)\right] \geq 1\,,$$

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(2) Deformation of $\mathcal{C}$ from $\mathrm{Im}(\sigma) = \frac{2m}{\varepsilon}$ to $\mathrm{Im}(\sigma) \to \infty$ does not get contribution from poles of $\widetilde{\Phi}_k^{-1}$ if and only if $(n, \ell, m) \in B$ where $B \subset \mathbb{Z}^3$ is a set of triples (n, ℓ, m) with $n, m > 0, 4mn - N\ell^2 > 0$ and

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Refinement Of DMZ Result

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- Another such example is $(n, \ell, m) = (13, 19, 7)$.

Future Directions

Open Questions

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- Positivity of degeneracy: the degeneracy is given by Fourier coefficient of interesting number theoretic object. Physics predicts the positivity of these Fourier coefficients [Sen (2010)]. Prove this mathematically. Proved for $\mathcal{M} = K3, N = 1$ [Bringmann-Murthy (2017), Rosselló (2024)].

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- The degeneracy of single-centered black holes is not captured by mock Jacobi forms for all charges. Formulate the notion of "mock Siegel forms" which captures the degeneracy for all charges [Bhand-Sen-RKS work in progress].

Thank You